

WPR - 3

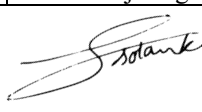
AMITY SCHOOL OF ENGINEERING & TECHNOLOGY
B.Tech (CSE - AIML) VII Semester

Project Area -- Generative AI & AI Safety (Trustworthy AI / MLOps)

Students Weekly Progress Report (WPR — Week 3) For Odd Semester of session 2026-2027

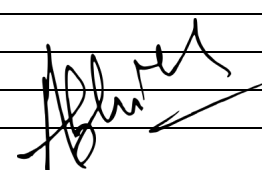
To be filled by Students	Group No. 50
Students Name	1. Shourya Solanki 2. Rachit Ryan Chug 3. Dhruv Raj Singh
Roll no.	1. 569 2. 166 3. 191
Enrollment no.	1. A2305223569 2. A2305223166 3. A2305223191
Project Title finalized, if Yes, give name, if NO, give reason	Yes. Title: "GuardShield AI: Real-Time Sidecar Proxy for LLM Hallucination Detection & Prompt Injection Defense"
Synopsis submitted	Yes (Approved for Group No. 50)
Literature review	Applied to synthetic dataset design. Incorporated Base64, ROT13, and Hinglish adversarial obfuscation attack vectors.
Technical & Economical Feasibility	Feasible. Python 3.11+, PyTorch CUDA (RTX 3050 GPU), DeBERTa-v3-small (86M SLM), Hugging Face Trainer. Zero monetary cost (₹0).
Bill of Material	NA
Project Progress Schedule (PERT Chart)	• W1-W3: Setup, open-source dataset ingestion & GuardShield-Bench-v1 synthetic creation (Completed / Week 3) • W4-W7: DeBERTa-v3 model fine-tuning & preprocessing pipeline • W8-W12: Token logit entropy algorithm, benchmark evaluation & IEEE paper draft
Design of critical components	1. Obfuscated Synthetic Dataset Generator (Base64/ROT13/Hinglish) 2. Dataset Splitting Pipeline (80% Train, 10% Val, 10% Test) 3. Pre-Scan DeBERTa-v3 Multi-Class Classifier Initialization
Fabrication work (give %)	0% (Software Project)
Experimental work (give %)	35% (Synthetic dataset generation, 401 records created, 80/10/10 split prepared, DeBERTa-v3 GPU initialization verified)
Result and Analysis	Generated GuardShield-Bench-v1 (401 synthetic obfuscated records); created train (320), val (40), test (41) splits; verified DeBERTa-v3 forward-pass on CUDA GPU.
Report writing	WPR Week 3 drafted.

Signature of students	1. Shourya Solanki Dhruv Raj Singh	2. Rachit Ryan Chug	3.
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Work done in this week (Week 3):

1. GuardShield-Bench-v1 Synthetic Dataset Creation: Built `obfuscation\_generator.py` applying Base64, ROT13, and Hinglish code-mixing to benchmark prompt injections, producing 401 standardized records.
2. Train/Val/Test Dataset Splitting: Implemented `prepare\_splits.py` dividing dataset into 80% Train (320 items), 10% Validation (40 items), and 10% Test (41 items) under `data/processed/splits/`.
3. DeBERTa-v3 Model Initialization: Developed `model\_init.py` initializing Hugging Face `microsoft/deberta-v3-small` (86M parameters) on local NVIDIA RTX 3050 CUDA GPU.
4. Baseline Forward-Pass Verification: Tested model tokenizer and classification logits, confirming pipeline readiness for fine-tuning.

To be filled by Guide (strike off whichever is not applicable)	
Performance of students is satisfactory	
Performance of students is unsatisfactory	
A warning to be issued to student(s) (Name)	
Student was not well (Name)	
Date 12/08/2026	 Signature of Guide